

Camera Class Reference

The Camera class represents a perspective camera.

```
#include <camera.h>
```

Public Member Functions

void	init	(const Box &b)
QMatrix4x4	viewMatrix	() const
QMatrix4x4	projectionMatrix	() const
Point	getObs	() const
void	updateClippingPlanes	(const Box &)
void	incrementDistance	(float inc)
void	setDistance	(float value)
void	incrementAngleX	(float inc)
void	setAngleX	(float value)
void	incrementAngleY	(float inc)
void	setAngleY	(float value)
void	setAngleZ	(float value)
Vector	eulerAngles	() const
void	pan	(const Vector &offset)
float	fov	() const
void	setFov	(float fovy)
void	setAspectRatio	(float ar)
float	aspectRatio	() const
float	znear	() const
void	setZnear	(float z)
float	zfar	() const
void	setZfar	(float z)

Member Function Documentation

float Camera::aspectRatio () const

Returns the camera aspect ratio.

Vector Camera::eulerAngles () const

Returns $v=(x,y,z)$ with the Euler angles of the camera. Rotation part of the viewing matrix is $\text{Rotz}(-v.z)*\text{Rotx}(v.x)*\text{Roty}(-v.y)$

float Camera::fov () const

Returns the vertical field of view (degrees).

Point Camera::getObs () const

Returns the camera (eye) position.

void Camera::incrementAngleX (float inc)

Rotates the camera by increasing the rotation angle around X axis (pitch).

void Camera::incrementAngleY (float inc)

Rotates the camera by increasing the rotation angle around Y axis (yaw).

void Camera::incrementDistance (float inc)

Moves the camera away ($\text{inc}>0$) or towards ($\text{inc}<0$) the current VRP.

void Camera::init (const Box & b)

Initializes the camera from a bounding box of the scene. The camera is placed outside the box, ensuring that the whole box fits within the viewing frustum.

void Camera::pan (const Vector & offset)

Offsets the camera position and VRP.

QMatrix4x4 Camera::projectionMatrix () const

Returns a 4x4 matrix representing the projection transform of the camera.

void Camera::setAngleX (float value)

Rotates the camera by setting the rotation angle around X axis (pitch).

void Camera::setAngleY (float value)

Rotates the camera by setting the rotation angle around Y axis (yaw).

void Camera::setAngleZ (float value)

Rotates the camera by setting the rotation angle around Z axis (roll).

void Camera::setAspectRatio (float ar)

Sets the camera aspect ratio.

void Camera::setDistance (float value)

Sets the distance to the current VRP.

void Camera::setFov (float fovy)

Sets the vertical field of view (degrees).

void Camera::setZfar (float z)

Sets Z far

void Camera::setZnear (float z)

Sets Z near

void Camera::updateClippingPlanes (const Box & box)

Recomputes the near, far clipping planes according to the given bounding box.

QMatrix4x4 Camera::viewMatrix () const

Returns a 4x4 matrix representing the viewing transform of the camera.

float Camera::zfar () const

Returns zfar (distance to the far clipping plane)

float Camera::znear () const

Returns znear (distance to the near clipping plane)